TOPIC: DEVELOPMENT OF A WEB-BASED WATER AND BEVERAGE FACTORY MANAGEMENT SYSTEM

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INTRODUCTION: The Water and Beverage Factory Management System (WBFMS) is a computerized solution designed to automate and integrate all major operations of a water and beverage factory including production, scheduling, stock management, order processing, employee management, and financial reporting. This system ensures transparency, speed, and accuracy across all departments.

PROBLEM STATEMENT: Currently, the factories face the following problems:

1. Difficulty in tracking daily production output and material usage.
2. Inconsistent production records leading to wastage and underproduction.
3. Payments are often made in cash without proper digital records.
4. Inaccurate stock updates due to unlinked sales and inventory systems.
5. Customers find it difficult to place orders easily because the factory lacks an online ordering system.
6. Customers cannot view available product types (e.g., bottled water sizes, beverages, or packages) in real time.
7. Customers are often unaware of the minimum quantity they can order, leading to order confusion.

These challenges often result in operational delays, financial losses, and reduced customer satisfaction**.**

AIM: To design and implement a computerized Water Factory Management System that automates production, sales, order management, and payment processes to enhance operational efficiency and customer service.

Users: Admin, Production Manager, Sales Manager, Customer

OBJECTIVE:

1. To develop a user-friendly ordering system where customers can view available products, order quantities, cancel or modify orders, and know the minimum order requirement.
2. To design a digital payment system that supports multiple payment methods, automates receipts, and synchronizes with sales and customer accounts.
3. To synchronize sales data with inventory and production modules for transparency.
4. To generate real-time reports for management decision-making and performance evaluation.
5. To reduce manual errors and improve overall customer satisfaction through digital operations.

REQUIRED TOOLS:

1. FRONTEND: HTML, CSS, JAVASCRIPT
2. BACKEND: PHP, PYTHON(Flask)
3. DATABASE: MySQL

CONCLUSION: The Water Factory Management System (with Beverages) aims to provide a complete digital transformation for water and beverage production companies. By integrating production, sales, payment, and delivery processes into one platform, the system will minimize operational errors, improve order fulfillment, and ensure better financial management.

Ultimately, this system will enhance efficiency, accuracy, and profitability while improving customer satisfaction and overall business performance.